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EDUCATION

The Hong Kong University of Science and Technology B.Eng. Sep. 2023 - Jun. 2027 (Expected)

• Integrative Systems & Design (Minor in Robotics) · CGA 3.43/4.3 · MCGA 3.52/4.3 · Rank 4/28

PUBLICATIONS & MANUSCRIPTS

ICML 2026 · Accepted

MVISTA-4D: View-Consistent 4D World Model with Test-Time Action Inference for Robotic Manipulation ^{arXiv}
Jiaxu Wang, Yicheng Jiang, **Tianlun He**, Jingkai Sun, Qiang Zhang, Junhao He, Jiahang Cao, Zesen Gan, Mingyuan Sun, Qiming Shao, Xiangyu Yue.

ICRA 2027 · Under Review

AR-WAM: A Visual-Conditioned Agent-Ready World Action Model for Robotic Manipulation

Yicheng Jiang, Zesen Gan, Xiaobo Wang, **Tianlun He**, Chenxu Zhao, Minghui Wu, Xinyue Wang, Jiaxu Wang, Junhao He, Jiannan Wang, Qiming Shao.

WindingWM: Winding Number-Guided Geometric World Model for Complex Assembly Planning

Yuhu Guo, **Tianlun He**, Yiming Li, Yue Chang, Yuhong Mo, Yuchao Wang, Fangzheng Zhong, Peter Yichen Chen.

WiSP: Winding-State Policies for Route-Class-Conditioned Robot Control

Yuhong Mo, **Tianlun He**, Yuhu Guo, Guoxin Fang, Yiming Li.

ECHO: Event-Augmented Context with Hindsight and Outlook for Wrist-Only Manipulation

Xinyue Wang, Yicheng Jiang, Zesen Gan, Junhao He, Jiaxu Wang, Jingtao Zhang, Junhao Li, **Tianlun He**, Jiannan Wang, Isabel Guan, Qiming Shao.

INTERNSHIP EXPERIENCE

Mondo Tech Robotics Algorithm Intern

May 2026 - Present

- **Robot Data Infrastructure:** Built the full Sonic teleoperation and data pipeline for an in-house humanoid and Unitree G1 + dexterous hands (capture, transfer, storage), powering a large-scale data-collection factory.
- **Data Standard & Platform:** Helped define the unified Mondo training-data format for multi-embodiment data; built a data-management platform with visualization, versioning and data-mix configuration.
- **World-Model Training & Evaluation:** Ran controlled comparisons across video-generative backbones (Cosmos, Wan2.2, GR00T) and data mixes, using open-loop and closed-loop G1 real-robot evaluation to guide selection.

RESEARCH EXPERIENCE

DexBotRL Real-World RL for Whole-Body Dexterous Manipulation

Jun. 2026 - Present

- Built a real-world RL system for whole-body dexterous manipulation on Unitree G1 + BrainCo Revo2 tactile hands (GR00T, RLinf); Actor-Critic improved success and robustness under limited real data, +22% success in simulation.
- Addressed sparse rewards, high human intervention and single-task generalization by exploring agent-driven dense rewards / multi-head critics, low-intervention online learning and a multi-task shared Actor-Critic.

Memory-Augmented VLA Undergraduate Research Program

Feb. 2026 - Jul. 2026

- Designed Memory VLA to mitigate memory forgetting in long-horizon manipulation, combining hierarchical task decomposition with a fixed-capacity compressive memory that preserves key history for long-range decisions.
- Led the VLA architecture design and iteration, including core components (subtask boundary detector, VAE compression); deployed and benchmarked on RMBench and VLABench, with ongoing memory-VLA literature review.

4D World Model for Robotic Manipulation ICML 2026 (Accepted)

Jan. 2026 - Feb. 2026

- Built an embodied 4D world model generating multi-view-consistent 3D/4D scene representations from single-view RGBD input, enabling an imagine-then-act paradigm for complex manipulation.
- Set up the real-robot platform (Piper arm + multi-view RGBD array + compute) and acquisition pipeline (multi-camera calibration, time sync, cross-view alignment); contributed to the 4D manipulation dataset.

PROJECT EXPERIENCE

Indoor Structural Semantic Segmentation Personal Research Project Dec. 2025 - Jan. 2026

- Designed a point-cloud semantic segmentation system for indoor structures, finely identifying ceilings, walls and other architectural elements, reaching 95% multi-class segmentation accuracy in real-world scenes.
- Built on Sonata pretraining + PTV3 backbone; owned annotation, dataset construction and training end-to-end.

Hydraulic Soft Robotic Arm Supervisor: Prof. Rob Scharff Aug. 2025 - Dec. 2025

- Designed and simulated the cable-driven mechanism and base structure for a hydraulic soft robotic arm.
- Contributed to embedded control and hardware integration, including hydraulic flow control and cable actuation.

HKUST RoboMaster Team ENTERPRIZE Head of Mechanical Dept. Sep. 2023 - Aug. 2025

- Designed core mechanics (serial-leg infantry — Best Infantry 2025, UAV, gimbal), all open-sourced.
- Led department management and cross-team coordination; built the design workflow and documentation system.

Bipedal RL Walking (IsaacGym) Supervisor: Dr. Lam Dec. 2024 - Jun. 2025

- Built a URDF-based reinforcement learning pipeline in IsaacGym, achieving stable bipedal walking in simulation.
- Built a motor torque test platform and learned a motor current-torque model with deep learning to prepare for sim-to-real transfer.

Personal Projects

- Tendon-Driven Dexterous Hand • Omnidirectional Wheelchair • 6-DoF Underwater Robot
- 3D-Printer Auto Material System (AMS) • RFID Smart Tool Cabinet • Desktop CNC Router

AWARDS

Scholarship: HKSAR Government Talent Development Scholarship (2024, 2025)

Competition: Champion — RoboMaster 2025 Univ. League & 2024 RMUC International

SKILLS

Embodied AI: VLA, world models, GR00T training/deployment; extensive real-robot and robotics R&D experience.

Engineering: Python, Linux, Git, full-stack embedded-to-algorithm; strong mechanical design (SolidWorks).